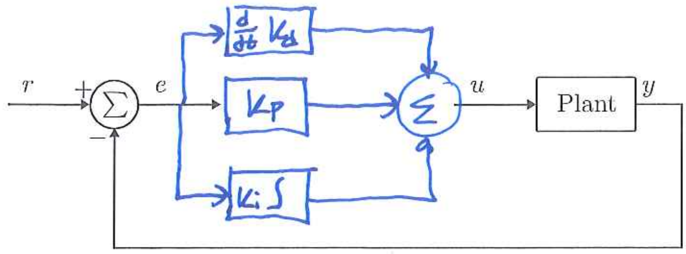
Test: Design of PID Controllers

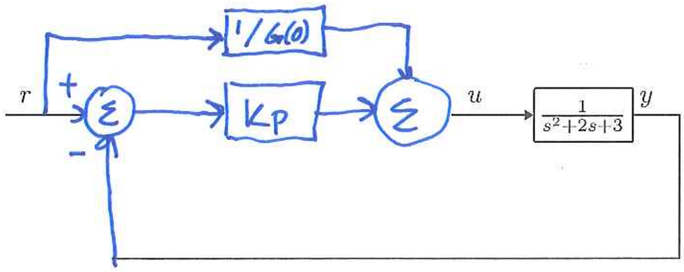
1. Include a PID-controller in the following block diagram.



2. Derive the closed-loop transfer function of a system $G(s) = \frac{1}{\tau s+1}$ with a P-control with gain K_p.

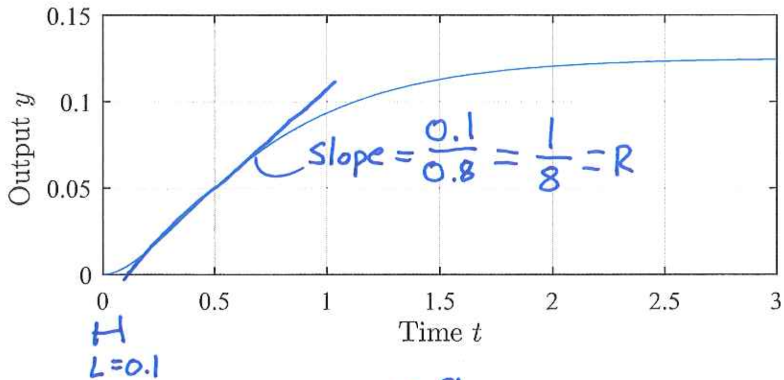
$$T(s) = \frac{k_p G(s)}{1 + k_p G(s)} = \frac{k_p/(\tau s+1)}{1 + k_p/(\tau s+1)} = \frac{k_p}{\tau s+1+k_p} = \frac{k_p/(1+k_p)}{\frac{\tau}{1+k_p} s + 1}$$

3. Include a P-controller with gain K_p in the following block diagram in addition to a feedforward, and compute an expression for the feedforward such that the steady state error is eliminated.



$G(0) = \frac{1}{3}$

4. Consider the following step response of an open-loop system. Use Ziegler-Nichols tuning method to provide initial guesses for a PI-controller for the system.



$K_p = \frac{0.9}{RL}$

$K_i = L/0.3$